

Method	$h = 5$ MSE	$h = 10$ MSE	$h = 10$ gain
Persistence	0.164028	0.226129	—
Autoregressive	0.149573 ± 0.000095	0.203950 ± 0.000504	—
Anchored additive	0.149606 ± 0.000224	0.200244 ± 0.000111	+1.817%
Spatial transport (ours)	0.145074 ± 0.000987	0.193141 ± 0.000991	+5.300%
Context-off	0.145143 ± 0.000954	0.193330 ± 0.001182	+5.208%
Action-free	0.149480 ± 0.000478	0.202249 ± 0.000865	+0.834%

Table 1: **Native 4×4 features.** All five arms share the 4×4 representation and support/query horizons; Context-off and Action-free explicitly remove information paths. Values are equal-seed means $\pm$ sample seed SD after within-episode window averaging; persistence is deterministic. Signed gain compares each arm with Autoregressive in the same column; green bold marks positive point estimates only. Paired uncertainty is reported separately. Native scores use shared-channel training normalization. All results use development validation.

Transport comparator	h	Gain	Difference [paired 95% interval]
Autoregressive	5	+3.008%	$-0.004499 [-0.006132, -0.002491]$
Autoregressive	10	+5.300%	$-0.010809 [-0.012461, -0.009028]$
Anchored additive	5	+3.029%	$-0.004532 [-0.005930, -0.002795]$
Anchored additive	10	+3.547%	$-0.007102 [-0.008831, -0.005019]$
Context-off	5	+0.048% [†]	$-0.000069 [-0.000633, +0.000478]$
Context-off	10	+0.097% [†]	$-0.000188 [-0.000943, +0.000485]$
Action-free	5	+2.948%	$-0.004406 [-0.006551, -0.002149]$
Action-free	10	+4.503%	$-0.009107 [-0.011896, -0.006061]$

Table 2: **Transport comparisons: Native 4×4 features.** Differences are Spatial transport (ours) minus comparator; negative favors ours. All four comparators and both endpoints are included. Intervals resample recording sessions and three training seeds jointly (10,000 draws; seed 173). [†] marks intervals containing zero. Green bold denotes positive point gain, not significance. Exploratory development evidence, without multiple-comparison adjustment.

Method	$h = 5$ MSE	$h = 10$ MSE	$h = 10$ gain
Persistence	0.163156	0.225622	—
Autoregressive	0.148577 ± 0.000135	0.202916 ± 0.000239	—
Anchored additive	0.148705 ± 0.000252	0.198704 ± 0.000131	+2.076%
Spatial transport (ours)	0.147921 ± 0.001069	0.197766 ± 0.001302	+2.538%
Context-off	0.147635 ± 0.000949	0.197553 ± 0.001423	+2.643%
Action-free	0.152414 ± 0.000690	0.206188 ± 0.000937	-1.613%

Table 3: **Original 2×2 coordinates.** All five arms share the 4×4 representation and support/query horizons; Context-off and Action-free explicitly remove information paths. Values are equal-seed means $\pm$ sample seed SD after within-episode window averaging; persistence is deterministic. Signed gain compares each arm with Autoregressive in the same column; green bold marks positive point estimates only. Paired uncertainty is reported separately. Predicted raw grids are pooled, then scored against exact original cached targets with original normalization. These values cannot be compared numerically with the native table. All results use development validation.

Transport comparator	h	Gain	Difference [paired 95% interval]
Autoregressive	5	+0.441% [†]	-0.000656 [-0.002499, +0.001918]
Autoregressive	10	+2.538%	-0.005150 [-0.007196, -0.002727]
Anchored additive	5	+0.527% [†]	-0.000784 [-0.002281, +0.001370]
Anchored additive	10	+0.472% [†]	-0.000938 [-0.002897, +0.001784]
Context-off	5	-0.194% [†]	+0.000286 [-0.000386, +0.001016]
Context-off	10	-0.108% [†]	+0.000213 [-0.000644, +0.001075]
Action-free	5	+2.948%	-0.004493 [-0.007086, -0.001778]
Action-free	10	+4.085%	-0.008422 [-0.011892, -0.004691]

Table 4: **Transport comparisons: Original 2×2 coordinates.** Differences are Spatial transport (ours) minus comparator; negative favors ours. All four comparators and both endpoints are included. Intervals resample recording sessions and three training seeds jointly (10,000 draws; seed 173). [†] marks intervals containing zero. Green bold denotes positive point gain, not significance. Exploratory development evidence, without multiple-comparison adjustment.

Method	Selected epochs	Total params	Active params
Autoregressive	16/25/22	1,026,305	1,007,776
Anchored additive	15/25/16	1,026,305	1,007,776
Spatial transport (ours)	14/19/11	1,026,305	1,026,305
Context-off	10/17/14	1,026,305	878,497
Action-free	6/5/4	1,026,305	988,001

Table 5: **All fifteen completed spatial models.** Every model completed 30 epochs; selected epochs follow seeds 0/1/2 and minimize window-weighted all-ten validation error. Active counts exclude frozen unused heads. The shared trunk does not imply identical active parameter counts; Context-off and Action-free remove inputs and parameters. All selected models passed relocated CPU forecast parity.